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Source: *Mathematics of Computation*, Apr., 1998, Vol. 67, No. 222 (Apr., 1998), pp. 647-665

Published by: American Mathematical Society

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APPROXIMATION PROPERTIES OF MULTIVARIATE WAVELETS

RONG-QING JIA

ABSTRACT. Wavelets are generated from refinable functions by using multiresolution analysis. In this paper we investigate the approximation properties of multivariate refinable functions. We give a characterization for the approximation order provided by a refinable function in terms of the order of the sum rules satisfied by the refinement mask. We connect the approximation properties of a refinable function with the spectral properties of the corresponding subdivision and transition operators. Finally, we demonstrate that a refinable function in $W_1^{k-1}(\mathbb{R}^s)$ provides approximation order k .

1. INTRODUCTION

We are concerned with functional equations of the form

$$(1.1) \quad \phi = \sum_{\alpha \in \mathbb{Z}^s} a(\alpha) \phi(M \cdot -\alpha),$$

where ϕ is the unknown function defined on the s -dimensional Euclidean space $\mathbb{R}^s$, a is a finitely supported sequence on $\mathbb{Z}^s$, and M is an $s \times s$ integer matrix such that $\lim_{n \rightarrow \infty} M^{-n} = 0$. The equation (1.1) is called a **refinement equation**, and the matrix M is called a **dilation matrix**. Correspondingly, the sequence a is called the **refinement mask**. Any function satisfying a refinement equation is called a **refinable function**.

If a satisfies

$$(1.2) \quad \sum_{\alpha \in \mathbb{Z}^s} a(\alpha) = m := |\det M|,$$

then it is known that there exists a unique compactly supported distribution ϕ satisfying the refinement equation (1.1) subject to the condition $\hat{\phi}(0) = 1$. This distribution is said to be **the normalized solution** to the refinement equation with mask a . This fact was essentially proved by Cavaretta, Dahmen, and Micchelli in [7, Chap. 5] for the case in which the dilation matrix is 2 times the $s \times s$ identity matrix I . The same proof applies to the general refinement equation (1.1).

Wavelets are generated from refinable functions. In [20], Jia and Micchelli discussed how to construct multivariate wavelets from refinable functions associated

Received by the editor April 17, 1996.

1991 *Mathematics Subject Classification*. Primary 41A25, 41A63; Secondary 42C15, 65D15.

Key words and phrases. Refinement equations, refinable functions, wavelets, accuracy, approximation order, smoothness, subdivision operators, transition operators.

Supported in part by NSERC Canada under Grant OGP 121336.

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with a general dilation matrix. The approximation and smoothness properties of wavelets are determined by the corresponding refinable functions.

In [9], DeVore, Jawerth, and Popov established a basic theory for nonlinear approximation by wavelets. In their work, the refinement mask was required to be nonnegative. In [15], Jia extended their results and, in particular, removed the restriction of non-negativity of the mask.

Our goal is to characterize the approximation order provided by a refinable function in terms of the refinement mask. This information is important for our understanding of wavelet approximation.

Before proceeding further, we introduce some notation. A multi-index is an s -tuple $\mu = (\mu_1, \dots, \mu_s)$ with its components being nonnegative integers. The length of μ is $|\mu| := \mu_1 + \dots + \mu_s$, and the factorial of μ is $\mu! := \mu_1! \cdots \mu_s!$. For two multi-indices $\mu = (\mu_1, \dots, \mu_s)$ and $\nu = (\nu_1, \dots, \nu_s)$, we write $\nu \leq \mu$ if $\nu_j \leq \mu_j$ for $j = 1, \dots, s$. If $\nu \leq \mu$, then we define

$$\binom{\mu}{\nu} := \frac{\mu!}{\nu!(\mu - \nu)!}.$$

For $j = 1, \dots, s$, D_j denotes the partial derivative with respect to the j th coordinate. For $\mu = (\mu_1, \dots, \mu_s)$, D^μ is the differential operator $D_1^{\mu_1} \cdots D_s^{\mu_s}$. Moreover, p_μ denotes the monomial given by

$$p_\mu(x) := x_1^{\mu_1} \cdots x_s^{\mu_s}, \quad x = (x_1, \dots, x_s) \in \mathbb{R}^s.$$

The total degree of p_μ is $|\mu|$. For a nonnegative integer k , we denote by Π_k the linear span of $\{p_\mu : |\mu| \leq k\}$. Then $\Pi := \bigcup_{k=0}^\infty \Pi_k$ is the linear space of all polynomials of s variables. We agree that $\Pi_{-1} = \{0\}$.

The Fourier transform of an integrable function f on $\mathbb{R}^s$ is defined by

$$\hat{f}(\xi) = \int_{\mathbb{R}^s} f(x) e^{-ix \cdot \xi} dx, \quad \xi \in \mathbb{R}^s,$$

where $x \cdot \xi$ denotes the inner product of two vectors x and ξ in $\mathbb{R}^s$. The domain of the Fourier transform can be naturally extended to include compactly supported distributions.

We denote by $\ell(\mathbb{Z}^s)$ the linear space of all sequences on $\mathbb{Z}^s$, and by $\ell_0(\mathbb{Z}^s)$ the linear space of all finitely supported sequences on $\mathbb{Z}^s$. For $\alpha \in \mathbb{Z}^s$, we denote by δ_α the element in $\ell_0(\mathbb{Z}^s)$ given by $\delta_\alpha(\alpha) = 1$ and $\delta_\alpha(\beta) = 0$ for all $\beta \in \mathbb{Z}^s \setminus \{\alpha\}$. In particular, we write δ for δ_0 . For $j = 1, \dots, s$, let e_j be the j th coordinate unit vector. The difference operator ∇_j on $\ell(\mathbb{Z}^s)$ is defined by $\nabla_j a := a - a(\cdot - e_j)$, $a \in \ell(\mathbb{Z}^s)$. For a multi-index $\mu = (\mu_1, \dots, \mu_s)$, ∇^μ is the difference operator $\nabla_1^{\mu_1} \cdots \nabla_s^{\mu_s}$.

For a compactly supported distribution ϕ on $\mathbb{R}^s$ and a sequence $b \in \ell(\mathbb{Z}^s)$, the **semi-convolution** of ϕ with b is defined by

$$\phi *' b := \sum_{\alpha \in \mathbb{Z}^s} \phi(\cdot - \alpha) b(\alpha).$$

Let $\mathbb{S}(\phi)$ denote the linear space $\{\phi *' b : b \in \ell(\mathbb{Z}^s)\}$. We call $\mathbb{S}(\phi)$ the **shift-invariant space** generated by ϕ . More generally, if Φ is a finite collection of compactly supported distributions on $\mathbb{R}^s$, then we use $\mathbb{S}(\Phi)$ to denote the linear space of all distributions of the form $\sum_{\phi \in \Phi} \phi *' b_\phi$, where $b_\phi \in \ell(\mathbb{Z}^s)$ for $\phi \in \Phi$.

Here is a brief outline of the paper. In Section 2 we clarify the relationship between the order of approximation provided by $\mathbb{S}(\phi)$ and the accuracy of ϕ , the

order of the polynomial space contained in $\mathbb{S}(\phi)$. In Section 3 we introduce the so-called sum rules and give a characterization for the accuracy of a refinable function in terms of the order of the sum rules satisfied by the refinement mask. In Section 4, several examples are provided to illustrate the general theory. Section 5 is devoted to a study of the subdivision and transition operators and their applications to approximation properties of refinable functions. Finally, in Section 6, we show that a refinable function in $W_1^k(\mathbb{R}^s)$ associated with an isotropic dilation matrix has accuracy at least $k + 1$.

2. APPROXIMATION ORDER AND POLYNOMIAL REPRODUCIBILITY

Let ϕ be a compactly supported function in $L_p(\mathbb{R}^s)$ ($1 \leq p \leq \infty$). In this section we clarify the relationship between the order of approximation provided by $\mathbb{S}(\phi)$ and the degree of the polynomial space contained in $\mathbb{S}(\phi)$. The reader is referred to [17] for a recent survey on approximation by shift-invariant spaces.

The norm in $L_p(\mathbb{R}^s)$ is denoted by $\|\cdot\|_p$. For an element $f \in L_p(\mathbb{R}^s)$ and a subset G of $L_p(\mathbb{R}^s)$, the distance from f to G , denoted by $\text{dist}_p(f, G)$, is defined by

$$\text{dist}_p(f, G) := \inf_{g \in G} \|f - g\|_p.$$

Let $S := \mathbb{S}(\phi) \cap L_p(\mathbb{R}^s)$. For $h > 0$, let $S^h := \{g(\cdot/h) : g \in S\}$. For a real number $\kappa \geq 0$, we say that $\mathbb{S}(\phi)$ provides **approximation order** κ if for each sufficiently smooth function f in $L_p(\mathbb{R}^s)$, there exists a constant $C > 0$ such that

$$\text{dist}_p(f, S^h) \leq C h^\kappa \quad \forall h > 0.$$

We say that $\mathbb{S}(\phi)$ provides **density order** κ (see [3]) if for each sufficiently smooth function f in $L_p(\mathbb{R}^s)$,

$$\lim_{h \rightarrow 0} \text{dist}_p(f, S^h)/h^\kappa = 0.$$

Let k be a positive integer. Suppose $\mathbb{S}(\phi) \supset \Pi_{k-1}$. Does $\mathbb{S}(\phi)$ always provide approximation order k ? The answer is a surprising no. The first counterexample was given by de Boor and Höllig in [4] by considering bivariate C^1 -cubics. Their results can be described in terms of box splines.

For a comprehensive study of box splines, the reader is referred to the book [5] by de Boor, Höllig, and Riemenschneider. For our purpose, it suffices to consider the box splines $M_{r,s,t}$ given by

$$\widehat{M}_{r,s,t}(\xi) = \left(\frac{1 - e^{-i\xi_1}}{i\xi_1} \right)^r \left(\frac{1 - e^{-i\xi_2}}{i\xi_2} \right)^s \left(\frac{1 - e^{-i(\xi_1 + \xi_2)}}{i(\xi_1 + \xi_2)} \right)^t, \quad \xi = (\xi_1, \xi_2) \in \mathbb{R}^2,$$

where r, s , and t are nonnegative integers. It is easily seen that $M_{r,s,t} \in L_\infty(\mathbb{R}^2)$ if and only if $\min\{r + s, s + t, t + r\} \geq 1$. Let $\phi_1 := M_{2,1,2}$ and $\phi_2 := M_{1,2,2}$. In [4], de Boor and Höllig proved that $\mathbb{S}(\phi_1, \phi_2) \supseteq \Pi_3$ but $\mathbb{S}(\phi_1, \phi_2)$ does not provide L_∞ -approximation order 4. In fact, the optimal L_∞ -approximation order provided by $\mathbb{S}(\phi_1, \phi_2)$ is 3. In [21], Ron showed that there exists a compactly supported function ψ in $\mathbb{S}(\phi_1, \phi_2)$ such that $\Pi_3 \subseteq \mathbb{S}(\psi)$. Since $\mathbb{S}(\psi) \subseteq \mathbb{S}(\phi_1, \phi_2)$, the approximation order provided by $\mathbb{S}(\psi)$ is at most 3.

In [6], de Boor and Jia extended the results in [4] in the following way. For $\rho = 1, 2, \dots$, let k be an integer such that $2\rho + 2 \leq k \leq 3\rho + 1$. Let

$$\Phi := \{M_{r,s,t} \in C^\rho(\mathbb{R}^2) : r + s + t \leq k + 2\}.$$

Then $\mathbb{S}(\Phi) \supset \Pi_k$, but the optimal L_p -approximation order ($1 \leq p \leq \infty$) provided by $\mathbb{S}(\Phi)$ is k , not $k + 1$.

However, if $\mathbb{S}(\phi)$ provides approximation order k , then $\mathbb{S}(\phi)$ contains Π_{k-1} . This was proved by Jia in [16]. Under the additional condition that $\hat{\phi}(0) \neq 0$, it was proved by Ron [21] that $\mathbb{S}(\phi)$ provides L_∞ -approximation order k if and only if $\mathbb{S}(\phi)$ contains Π_{k-1} . In general, we have the following results, which were established in [16].

Theorem 2.1. *Let $1 \leq p \leq \infty$, and let ϕ be a compactly supported function in $L_p(\mathbb{R}^s)$ with $\hat{\phi}(0) \neq 0$. For every positive integer k , the following statements are equivalent:*

- (a) $\mathbb{S}(\phi)$ provides approximation order k .
- (b) $\mathbb{S}(\phi)$ provides density order $k - 1$.
- (c) $\mathbb{S}(\phi)$ contains Π_{k-1} .
- (d) $D^\mu \hat{\phi}(2\pi\beta) = 0$ for all μ with $|\mu| \leq k - 1$ and all $\beta \in \mathbb{Z}^s \setminus \{0\}$.

We remark that the implications (a) $\Rightarrow$ (b) $\Rightarrow$ (c) $\Rightarrow$ (d) are valid without the assumption $\hat{\phi}(0) \neq 0$. Indeed, (a) $\Rightarrow$ (b) is obvious, (b) $\Rightarrow$ (c) was proved in [16], and the implication (c) $\Rightarrow$ (d) was established in [2].

Suppose ϕ is the normalized solution of the refinement equation (1.1). If ϕ lies in $L_p(\mathbb{R}^s)$ for some p , $1 \leq p \leq \infty$, then Theorem 2.1 applies to ϕ , because $\hat{\phi}(0) = 1$. Thus, there are two questions of interest. The first question is how to determine whether ϕ lies in $L_p(\mathbb{R}^s)$, and the second problem is how to characterize the highest degree of polynomials contained in $\mathbb{S}(\phi)$. The first question was discussed by Han and Jia in [12]. In this paper, we concentrate on the second question. When we speak of polynomial containment, ϕ is not required to be an integrable function. Thus, we say that a compactly supported distribution ϕ on $\mathbb{R}^s$ has **accuracy** k , if $\mathbb{S}(\phi) \supset \Pi_{k-1}$ (see [13] for the terminology of accuracy).

We point out that the equivalence between (c) and (d) in Theorem 2.1 remains true for every compactly supported distribution ϕ on $\mathbb{R}^s$.

If ϕ is a compactly supported continuous function on $\mathbb{R}^s$, and if ϕ satisfies condition (d), then it was proved in [14] that

$$(2.1) \qquad \phi *' p = \hat{\phi}(-iD) p \qquad \forall p \in \Pi_{k-1},$$

where i is the imaginary unit and $\hat{\phi}(-iD)$ denotes the differential operator given by the formal power series

$$\sum_{\mu \geq 0} \frac{D^\mu \hat{\phi}(0)}{\mu!} (-iD)^\mu.$$

For a given polynomial p , $D^\mu p = 0$ if $|\mu|$ is sufficiently large. Thus, $\hat{\phi}(-iD)$ is well defined on Π . We indicate that (2.1) is also valid for a compactly supported distribution ϕ on $\mathbb{R}^s$ satisfying condition (d). To see this, choose a function $\rho \in C_c^\infty(\mathbb{R}^s)$ such that $\hat{\rho}(0) = 1$ and $D^\nu \hat{\rho}(0) = 0$ for all ν with $0 < |\nu| \leq k - 1$. Let $\rho_n := \rho(\cdot/n)/n^s$ for $n = 1, 2, \dots$. Then for each n , $\phi_n := \phi * \rho_n$, the convolution of ϕ with ρ_n , is a function in $C_c^\infty(\mathbb{R}^s)$. Moreover, the sequence $(\phi_n)_{n=1,2,\dots}$ converges to ϕ in the sense that

$$\lim_{n \rightarrow \infty} \langle \phi_n, f \rangle = \langle \phi, f \rangle \qquad \forall f \in C_c^\infty(\mathbb{R}^s).$$

See [1, p. 97] for these facts. Thus, we have $\hat{\phi}_n(\xi) = \hat{\phi}(\xi)\hat{\rho}_n(\xi)$ for $\xi \in \mathbb{R}^s$. Since ϕ satisfies condition (d), by using the Leibniz formula for differentiation, we get $D^\mu \hat{\phi}_n(2\pi\beta) = 0$ for $|\mu| \leq k-1$ and $\beta \in \mathbb{Z}^s \setminus \{0\}$. Hence (2.1) is applicable to ϕ_n and

$$\phi_n *' p = \hat{\phi}_n(-iD)p \quad \forall p \in \Pi_{k-1}.$$

Letting $n \rightarrow \infty$ in the above equation, we obtain $\phi *' p = \hat{\phi}(-iD)p$ for all $p \in \Pi_{k-1}$. Consequently, the linear mapping $\phi *'$ given by $p \mapsto \phi *' p$ maps Π_{k-1} to Π_{k-1} . If, in addition, $\hat{\phi}(0) \neq 0$, then this mapping is one-to-one, and hence it is onto. This shows that (d) $\Rightarrow$ (c) is valid for every compactly supported distribution ϕ on $\mathbb{R}^s$ with $\hat{\phi}(0) \neq 0$.

Next, we show that (c) $\Rightarrow$ (d) for every compactly supported distribution ϕ on $\mathbb{R}^s$. If ϕ is a compactly supported continuous function on $\mathbb{R}^s$, this was proved in [2] and [14]. Let ϕ be a compactly supported distribution on $\mathbb{R}^s$. For a fixed element $\beta \in \mathbb{Z}^s \setminus \{0\}$, choose a function $\rho \in C_c^\infty(\mathbb{R}^s)$ such that $\hat{\rho}(0) \neq 0$ and $\hat{\rho}(2\pi\beta) \neq 0$. Then the convolution $\phi * \rho$ is a function in $C_c^\infty(\mathbb{R}^s)$ and its Fourier transform is $\hat{\phi}\hat{\rho}$. Note that the mapping $\rho *'$ given by $q \mapsto \rho * q$ maps Π_{k-1} to Π_{k-1} . Since $\hat{\rho}(0) \neq 0$, this mapping is one-to-one; hence it is onto. Thus, for $p \in \Pi_{k-1}$, we can find $q \in \Pi_{k-1}$ such that $p = \rho * q$. Since $\mathbb{S}(\phi) \supset \Pi_{k-1}$, there exists some $b \in \ell(\mathbb{Z}^s)$ such that $q = \phi *' b$. It follows that $p = \rho * (\phi *' b) = (\rho * \phi) *' b$. This shows that $\mathbb{S}(\phi * \rho) \supset \Pi_{k-1}$. By what has been proved, $D^\mu (\hat{\phi}\hat{\rho})(2\pi\beta) = 0$ for all μ with $|\mu| \leq k-1$. Since $\hat{\rho}(2\pi\beta) \neq 0$, we can write $\hat{\phi} = (\hat{\phi}\hat{\rho})(1/\hat{\rho})$ in a neighborhood of $2\pi\beta$. By applying the Leibniz formula for differentiation to this equation, we obtain $D^\mu \hat{\phi}(2\pi\beta) = 0$ for $|\mu| \leq k-1$. This shows that (c) $\Rightarrow$ (d) for every compactly supported distribution ϕ on $\mathbb{R}^s$.

To summarize, a compactly supported distribution ϕ on $\mathbb{R}^s$ with $\hat{\phi}(0) \neq 0$ possesses accuracy k if and only if $D^\mu \hat{\phi}(2\pi\beta) = 0$ for all μ with $|\mu| \leq k-1$ and all $\beta \in \mathbb{Z}^s \setminus \{0\}$.

3. CHARACTERIZATION OF ACCURACY

The purpose of this section is to give a characterization for the accuracy of a refinable function in terms of the refinement mask.

For an $s \times s$ dilation matrix M , let Γ be a complete set of representatives of the distinct cosets of $\mathbb{Z}^s/M\mathbb{Z}^s$, and let Ω be a complete set of representatives of the distinct cosets of $\mathbb{Z}^s/M^T\mathbb{Z}^s$, where M^T denotes the transpose of M . Evidently, $\#\Gamma = \#\Omega = |\det M|$. Without loss of any generality, we may assume that $0 \in \Gamma$ and $0 \in \Omega$.

Suppose a is a finitely supported sequence on $\mathbb{Z}^s$ satisfying (1.2). Let ϕ be the normalized solution of the refinement equation (1.1). Taking Fourier transform of both sides of (1.1), we obtain

$$(3.1) \quad \hat{\phi}(\xi) = H((M^T)^{-1}\xi) \hat{\phi}((M^T)^{-1}\xi), \quad \xi \in \mathbb{R}^s,$$

where

$$(3.2) \quad H(\xi) := \sum_{\alpha \in \mathbb{Z}^s} a(\alpha) e^{-i\alpha \cdot \xi} / m, \quad \xi \in \mathbb{R}^s.$$

Note that H is a 2π -periodic function and $H(0) = 1$.

For a compactly supported distribution ϕ on $\mathbb{R}^s$, define

$$N(\phi) := \{\xi \in \mathbb{R}^s : \hat{\phi}(\xi + 2\pi\beta) = 0 \ \forall \beta \in \mathbb{Z}^s\}.$$

If ϕ is a compactly supported function in $L_p(\mathbb{R}^s)$ ($1 \leq p \leq \infty$), then the shifts of ϕ are stable if and only if $N(\phi)$ is the empty set (see [19]).

Theorem 3.1. *Let a be a finitely supported sequence on $\mathbb{Z}^s$ satisfying (1.2), and let H be the function given in (3.2). If*

$$(3.3) \quad D^\mu H(2\pi(M^T)^{-1}\omega) = 0 \quad \forall \omega \in \Omega \setminus \{0\} \text{ and } |\mu| \leq k-1,$$

then the normalized solution ϕ of the refinement equation (1.1) has accuracy k . Conversely, if ϕ has accuracy k , and if $N(\phi) \cap (2\pi(M^T)^{-1}\Omega) = \emptyset$, then (3.3) holds true.

Proof. Suppose that (3.3) is satisfied. Since H is 2π -periodic, (3.3) implies

$$(3.4) \quad D^\mu H(2\pi(M^T)^{-1}\beta) = 0 \quad \forall \beta \in \mathbb{Z}^s \setminus (M^T\mathbb{Z}^s) \text{ and } |\mu| \leq k-1.$$

Let f and g be the functions given by

$$f(\xi) := H((M^T)^{-1}\xi) \quad \text{and} \quad g(\xi) := \hat{\phi}((M^T)^{-1}\xi), \quad \xi \in \mathbb{R}^s.$$

For $|\mu| \leq k-1$ and $\beta \in \mathbb{Z}^s \setminus \{0\}$, applying the Leibniz formula for differentiation to (3.1), we obtain

$$(3.5) \quad D^\mu \hat{\phi}(2\pi\beta) = \sum_{\nu \leq \mu} \binom{\mu}{\nu} D^\nu f(2\pi\beta) D^{\mu-\nu} g(2\pi\beta).$$

By using the chain rule, we see that $D^\nu f(2\pi\beta)$ is a linear combination of terms of the form $D^\alpha H(2\pi(M^T)^{-1}\beta)$, where $\alpha \leq \nu$. In light of (3.4), these terms are equal to 0 if $\beta \in \mathbb{Z}^s \setminus (M^T\mathbb{Z}^s)$. This shows that $D^\mu \hat{\phi}(2\pi\beta) = 0$ for $\beta \in \mathbb{Z}^s \setminus (M^T\mathbb{Z}^s)$.

We shall prove that, for $r = 0, 1, \dots$, $D^\mu \hat{\phi}(2\pi\beta) = 0$ for $\beta \in ((M^T)^r\mathbb{Z}^s) \setminus ((M^T)^{r+1}\mathbb{Z}^s)$. This will be done by induction on r . The case $r = 0$ was established above. Suppose $r \geq 1$ and our claim has been verified for $r-1$. Let $\beta \in ((M^T)^r\mathbb{Z}^s) \setminus ((M^T)^{r+1}\mathbb{Z}^s)$. Then we have $(M^T)^{-1}\beta \in ((M^T)^{r-1}\mathbb{Z}^s) \setminus ((M^T)^r\mathbb{Z}^s)$. Hence, by the induction hypothesis, $D^\mu \hat{\phi}(2\pi(M^T)^{-1}\beta) = 0$ for $|\mu| \leq k-1$. Consequently, $D^\mu g(2\pi\beta) = 0$ for all μ with $|\mu| \leq k-1$. This in connection with (3.5) tells us that $D^\mu \hat{\phi}(2\pi\beta) = 0$ for $|\mu| \leq k-1$, thereby completing the induction procedure. The sufficiency part of the theorem has been established.

Conversely, suppose ϕ has accuracy k and $N(\phi) \cap (2\pi(M^T)^{-1}\Omega) = \emptyset$. Then

$$D^\mu \hat{\phi}(2\pi\beta) = 0 \quad \forall \beta \in \mathbb{Z}^s \setminus \{0\} \text{ and } |\mu| \leq k-1.$$

Let $\omega \in \Omega \setminus \{0\}$. Since $N(\phi) \cap (2\pi(M^T)^{-1}\Omega) = \emptyset$, there exists some $\beta \in \mathbb{Z}^s$ such that $\hat{\phi}(\gamma) \neq 0$ for $\gamma := 2\pi\beta + 2\pi(M^T)^{-1}\omega$. Thus, the following identity is valid for ξ in a neighborhood of γ :

$$H(\xi) = \hat{\phi}(M^T\xi)[1/\hat{\phi}(\xi)].$$

Let h be the function given by $\xi \mapsto \hat{\phi}(M^T\xi)$, $\xi \in \mathbb{R}^s$. By using the Leibniz formula for differentiation, we obtain

$$D^\mu H(\gamma) = \sum_{\nu \leq \mu} \binom{\mu}{\nu} D^\nu h(\gamma) D^{\mu-\nu}[1/\hat{\phi}](\gamma).$$

By the chain rule, $D^\nu h(\gamma)$ is a linear combination of terms of the form $D^\alpha \hat{\phi}(M^T \gamma)$, where $\alpha \leq \nu$. Note that

$$M^T \gamma = M^T (2\pi \beta + 2\pi (M^T)^{-1} \omega) = 2\pi (M^T) \beta + 2\pi \omega \in 2\pi \mathbb{Z}^s \setminus \{0\}.$$

Hence $D^\alpha \hat{\phi}(M^T \gamma) = 0$ for $|\alpha| \leq k-1$, because ϕ has accuracy k . Therefore we obtain $D^\mu H(2\pi \beta + 2\pi (M^T)^{-1} \omega) = 0$ for $|\mu| \leq k-1$. But H is 2π -periodic. This shows that $D^\mu H(2\pi (M^T)^{-1} \omega) = 0$ for all $\omega \in \Omega \setminus \{0\}$ and $|\mu| \leq k-1$, as desired. The proof of the theorem is complete. $\square$

In the rest of this section we shall show that (3.3) is equivalent to saying that, for all $p \in \Pi_{k-1}$,

$$(3.6) \quad \sum_{\beta \in \mathbb{Z}^s} a(M\beta) p(M\beta) = \sum_{\beta \in \mathbb{Z}^s} a(M\beta + \gamma) p(M\beta + \gamma) \quad \forall \gamma \in \Gamma.$$

For this purpose, we first establish the following lemma.

Lemma 3.2. *The matrix*

$$(3.7) \quad \frac{1}{\sqrt{m}} (e^{i2\pi M^{-1}\gamma \cdot \omega})_{\gamma \in \Gamma, \omega \in \Omega}$$

is a unitary one.

Proof. Let $\gamma \in \Gamma \setminus \{0\}$. We claim that there exists some $\omega' \in \Omega$ such that $M^{-1}\gamma \cdot \omega' \notin \mathbb{Z}$. Any element $\beta \in \mathbb{Z}^s$ can be represented as $M^T \alpha + \omega$ for some $\alpha \in \mathbb{Z}^s$ and $\omega \in \Omega$. Note that $(M^{-1}\gamma) \cdot (M^T \alpha) = \gamma \cdot \alpha \in \mathbb{Z}$ for all $\alpha \in \mathbb{Z}^s$. Hence $M^{-1}\gamma \cdot \omega' \in \mathbb{Z}$ for all $\omega' \in \Omega$ implies that $M^{-1}\gamma \cdot \beta \in \mathbb{Z}$ for all $\beta \in \mathbb{Z}^s$. In other words, $M^{-1}\gamma \in \mathbb{Z}^s$, and hence $\gamma \in M\mathbb{Z}^s$, which contradicts the assumption $\gamma \in \Gamma \setminus \{0\}$. This verifies our claim.

For a fixed element γ in $\Gamma \setminus \{0\}$, let

$$\sigma := \sum_{\omega \in \Omega} e^{i2\pi M^{-1}\gamma \cdot \omega}.$$

Choose $\omega' \in \Omega$ such that $M^{-1}\gamma \cdot \omega' \notin \mathbb{Z}$. We have

$$e^{i2\pi M^{-1}\gamma \cdot \omega'} \sigma = \sum_{\omega \in \Omega} e^{i2\pi (M^{-1}\gamma) \cdot (\omega + \omega')} = \sum_{\omega \in \Omega} e^{i2\pi M^{-1}\gamma \cdot \omega} = \sigma.$$

Since $e^{i2\pi M^{-1}\gamma \cdot \omega'} \neq 1$, it follows that $\sigma = 0$. This shows that

$$(3.8) \quad \sum_{\omega \in \Omega} e^{i2\pi M^{-1}\gamma \cdot \omega} = 0 \quad \forall \gamma \in \Gamma \setminus \{0\}.$$

Similarly, we can prove that

$$(3.9) \quad \sum_{\gamma \in \Gamma} e^{i2\pi M^{-1}\gamma \cdot \omega} = 0 \quad \forall \omega \in \Omega \setminus \{0\}.$$

Finally, the matrix in (3.7) is unitary if and only if for every pair of elements $\gamma, \gamma' \in \Gamma$,

$$\frac{1}{m} \sum_{\omega \in \Omega} e^{i2\pi M^{-1}(\gamma - \gamma') \cdot \omega} = \begin{cases} 1 & \text{if } \gamma = \gamma', \\ 0 & \text{if } \gamma \neq \gamma'. \end{cases}$$

For $\gamma = \gamma'$, this comes from the fact $\#\Omega = m$; for $\gamma \neq \gamma'$, this follows from (3.8). $\square$

Lemma 3.3. *Let a be a finitely supported sequence satisfying (1.2), and let H be the function given in (3.2). Then the following two conditions are equivalent for every polynomial p :*

- (a) $p(iD)H(2\pi(M^T)^{-1}\omega) = 0$ for all $\omega \in \Omega \setminus \{0\}$.
 (b) $\sum_{\beta \in \mathbb{Z}^s} a(M\beta)p(M\beta) = \sum_{\beta \in \mathbb{Z}^s} a(M\beta + \gamma)p(M\beta + \gamma)$ for all $\gamma \in \Gamma$.

Proof. By (3.2) we have

$$mp(iD)H(\xi) = \sum_{\alpha \in \mathbb{Z}^s} a(\alpha)p(\alpha)e^{-i\alpha \cdot \xi}, \quad \xi \in \mathbb{R}^s.$$

An element $\alpha \in \mathbb{Z}^s$ can be written uniquely as $M\beta + \gamma$ with $\beta \in \mathbb{Z}^s$ and $\gamma \in \Gamma$. Observe that, for $\xi := 2\pi(M^T)^{-1}\omega$,

$$-i\alpha \cdot \xi = -i(M\beta + \gamma) \cdot 2\pi(M^T)^{-1}\omega = -i2\pi\beta \cdot \omega - i2\pi\gamma \cdot (M^T)^{-1}\omega.$$

Hence we have

$$(3.10) \quad mp(iD)H(2\pi(M^T)^{-1}\omega) = \sum_{\gamma \in \Gamma} b(\gamma)e^{-i2\pi\gamma \cdot (M^T)^{-1}\omega},$$

where

$$b(\gamma) := \sum_{\beta \in \mathbb{Z}^s} a(M\beta + \gamma)p(M\beta + \gamma).$$

Condition (b) says that $b(\gamma) = b(0)$ for all $\gamma \in \Gamma$. Hence by (3.9) we deduce from (3.10) that

$$mp(iD)H(2\pi(M^T)^{-1}\omega) = b(0) \sum_{\gamma \in \Gamma} e^{-i2\pi\gamma \cdot (M^T)^{-1}\omega} = 0$$

for all $\omega \in \Omega \setminus \{0\}$. This shows that (b) $\Rightarrow$ (a).

Conversely, (3.10) tells us that condition (a) implies

$$\sum_{\gamma \in \Gamma} b(\gamma)e^{-i2\pi M^{-1}\gamma \cdot \omega} = 0 \quad \forall \omega \in \Omega \setminus \{0\}.$$

Let η be an element of Γ . Then it follows that

$$\sum_{\omega \in \Omega} e^{i2\pi M^{-1}\eta \cdot \omega} \sum_{\gamma \in \Gamma} b(\gamma)e^{-i2\pi M^{-1}\gamma \cdot \omega} = \sum_{\gamma \in \Gamma} b(\gamma).$$

On the other hand,

$$\sum_{\omega \in \Omega} e^{i2\pi M^{-1}\eta \cdot \omega} \sum_{\gamma \in \Gamma} b(\gamma)e^{-i2\pi M^{-1}\gamma \cdot \omega} = \sum_{\gamma \in \Gamma} b(\gamma) \sum_{\omega \in \Omega} e^{i2\pi M^{-1}(\eta - \gamma) \cdot \omega} = mb(\eta),$$

since $\sum_{\omega \in \Omega} e^{i2\pi M^{-1}(\eta - \gamma) \cdot \omega} = 0$ for $\gamma \neq \eta$, by Lemma 3.2. This shows $mb(\eta) = \sum_{\gamma \in \Gamma} b(\gamma)$. Therefore $b(\eta) = b(0)$ for all $\eta \in \Gamma$. In other words, (a) implies (b). $\square$

If an element $a \in \ell_0(\mathbb{Z}^s)$ satisfies (3.6) for all $p \in \Pi_{k-1}$, then we say that a satisfies the **sum rules** of order k . The results of this section can be summarized as follows: If the refinement mask a satisfies the sum rules of order k , then the normalized solution ϕ of the refinement equation with mask a has accuracy k . Conversely, if ϕ has accuracy k , and if $N(\phi) \cap (2\pi(M^T)^{-1}\Omega) = \emptyset$, then a satisfies the sum rules of order k .

4. EXAMPLES

In this section we give several examples to illustrate the general theory. The **symbol** of a sequence $a \in \ell_0(\mathbb{Z}^s)$ is the Laurent polynomial $\tilde{a}(z)$ given by

$$\tilde{a}(z) := \sum_{\alpha \in \mathbb{Z}^s} a(\alpha) z^\alpha, \qquad z \in (\mathbb{C} \setminus \{0\})^s,$$

where $z^\alpha := z_1^{\alpha_1} \cdots z_s^{\alpha_s}$ for $z = (z_1, \dots, z_s) \in \mathbb{C}^s$ and $\alpha = (\alpha_1, \dots, \alpha_s) \in \mathbb{Z}^s$. If a is supported on $[0, N]^s$ for some positive integer N , then $\tilde{a}(z)$ is a polynomial of z .

In the univariate case ($s = 1$), if a satisfies the sum rules of order k , then $\tilde{a}(z)$ is divisible by $(1 + z)^k$ (see, e.g., [8]). In the multivariate case ($s > 1$), this is no longer true.

Example 4.1. Let $s = 2$ and $M = 2I$, where I is the 2×2 identity matrix. Let a be the sequence on $\mathbb{Z}^2$ given by its symbol

$$\tilde{a}(z) := z_1^2 + z_2 + z_1 z_2 + z_1 z_2^2.$$

Then a satisfies the sum rules of order 1. But the polynomial $\tilde{a}(z)$ is irreducible.

It is easy to verify that a satisfies the sum rules of order 1. Let us show that $\tilde{a}(z)$ is irreducible. Suppose to the contrary that $\tilde{a}(z)$ is reducible. Then $\tilde{a}(z)$ can be factored as

$$\tilde{a}(z) = f(z)g(z),$$

where f and g are polynomials of (total) degree at least 1. Since the degree of $\tilde{a}(z)$ is 3, the degree of either f or g is 1. Suppose the degree of f is 1 and

$$f(z_1, z_2) = \lambda z_1 + \mu z_2 + \nu,$$

where λ, μ, ν are complex numbers and either $\lambda \neq 0$ or $\mu \neq 0$. If $\lambda \neq 0$, then for all $z_2 \in \mathbb{C}$, $f(-(\mu z_2 + \nu)/\lambda, z_2) = 0$, and so

$$\tilde{a}(-(\mu z_2 + \nu)/\lambda, z_2) = 0 \qquad \forall z_2 \in \mathbb{C}.$$

If $\mu \neq 0$, then $\tilde{a}(-(\mu z_2 + \nu)/\lambda, z_2)$ is a polynomial of z_2 of degree 3 with $-\mu/\lambda$ being its leading coefficient. Hence $\mu = 0$. But it is also impossible that $\tilde{a}(-\nu/\lambda, z_2) = 0$ for all $z_2 \in \mathbb{C}$. Thus, we must have $\lambda = 0$, and hence $\tilde{a}(z_1, -\nu/\mu) = 0$ for all $z_1 \in \mathbb{C}$. However, $\tilde{a}(z_1, -\nu/\mu)$ is a polynomial of z_1 of degree 2 with 1 being its leading coefficient. This contradiction shows that $\tilde{a}(z)$ is irreducible.

Let a be the sequence given as above, and let ϕ be the normalized solution of the refinement equation

$$\phi = \sum_{\alpha \in \mathbb{Z}^2} a(\alpha) \phi(2 \cdot -\alpha).$$

Then ϕ lies in $L_2(\mathbb{R}^2)$. This can be verified by using the results in [12]. Let b be the element in $\ell_0(\mathbb{Z}^2)$ given by its symbol

$$\tilde{b}(z) := |\tilde{a}(z)|^2/4 \qquad \text{for } |z_1| = 1 \text{ and } |z_2| = 1.$$

We have

$$\begin{aligned} 4\tilde{b}(z) &= 4 + z_1 + z_1^{-1} + z_2 + z_2^{-1} + z_1 z_2 + z_1^{-1} z_2^{-1} \\ &\quad + z_1 z_2^{-1} + z_1^{-1} z_2 + z_1 z_2^{-2} + z_1^{-1} z_2^2 + z_1^2 z_2^{-1} + z_1^{-2} z_2. \end{aligned}$$

Let B be the linear operator on $\ell_0(\mathbb{Z}^2)$ given by

$$Bv(\alpha) := \sum_{\beta \in \mathbb{Z}^2} b(2\alpha - \beta) v(\beta), \quad \alpha \in \mathbb{Z}^2,$$

where $v \in \ell_0(\mathbb{Z}^2)$. Let W be the B -invariant subspace generated by $-\delta_{-e_1} + 2\delta - \delta_{e_1}$ and $-\delta_{-e_2} + 2\delta - \delta_{e_2}$. Then the spectral radius ρ of the linear operator $B|_W$ is $3/4$. Since $\rho < 1$, by [12, Theorems 3.3 and 4.1], the subdivision scheme associated with a is L_2 -convergent. Therefore, $\phi \in L_2(\mathbb{R}^2)$ and the shifts of ϕ are orthonormal (see [11]). We conclude that the optimal order of approximation provided by $\mathbb{S}(\phi)$ is 1.

If the refinement mask a satisfies the sum rules of order k , then the normalized solution ϕ of the refinement equation with mask a has accuracy k . However, if the condition $N(\phi) \cap (2\pi(M^T)^{-1}\Omega) = \emptyset$ is not satisfied, then ϕ could have higher accuracy. For instance, the function ϕ on $\mathbb{R}$ given by $\phi(x) = 1/2$ for $0 \leq x < 2$ and $\phi(x) = 0$ for $x \in \mathbb{R} \setminus [0, 2)$ satisfies the refinement equation

$$\phi = \sum_{\alpha \in \mathbb{Z}} a(\alpha) \phi(2 \cdot -\alpha),$$

where the symbol of the mask a is $\tilde{a}(z) = 1 + z^2$. Then a does not satisfy the sum rules of order 1. But ϕ has accuracy 1, and $\mathbb{S}(\phi)$ provides L_∞ -approximation order 1. The following is an example in the two-dimensional case.

Example 4.2. Let ϕ be the Zwart-Powell element defined by its Fourier transform

$$\hat{\phi}(\xi_1, \xi_2) := g(\xi_1) g(\xi_2) g(\xi_1 + \xi_2) g(-\xi_1 + \xi_2), \quad (\xi_1, \xi_2) \in \mathbb{R}^2,$$

where g is the function on $\mathbb{R}$ given by $\xi \mapsto (1 - e^{-i\xi})/(i\xi)$, $\xi \in \mathbb{R}$. Then ϕ is a compactly supported continuous function on $\mathbb{R}^2$ and $\mathbb{S}(\phi)$ provides L_∞ -approximation order 3. On the other hand, ϕ is refinable but the corresponding mask does not satisfy the sum rules of order 3.

For the first statement the reader is referred to [5, p. 72]. Let us verify the second statement. From [5, p. 140] we know that the Zwart-Powell element ϕ is refinable and the corresponding mask a is given by $a(\alpha) = 0$ for $\alpha \in \mathbb{Z}^2 \setminus [-1, 2] \times [0, 3]$ and

$$(a(\alpha_1, \alpha_2))_{-1 \leq \alpha_1 \leq 2, 0 \leq \alpha_2 \leq 3} = \frac{1}{4} \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 2 & 2 & 1 \\ 1 & 2 & 2 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}.$$

Evidently, the mask a satisfies the sum rules of order 2, but a does not satisfy the sum rules of order 3. Note that $(\pi, \pi) \in N(\phi)$ in this case.

Example 4.3. Let M be the matrix

$$\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix},$$

and let a be the sequence on $\mathbb{Z}^2$ such that $a(\alpha) = 0$ for $\alpha \in \mathbb{Z}^2 \setminus [-2, 2]^2$ and

$$(a(\alpha_1, \alpha_2))_{-2 \leq \alpha_1, \alpha_2 \leq 2} = \frac{1}{32} \begin{bmatrix} 0 & -1 & 0 & -1 & 0 \\ -1 & 0 & 10 & 0 & -1 \\ 0 & 10 & 32 & 10 & 0 \\ -1 & 0 & 10 & 0 & -1 \\ 0 & -1 & 0 & -1 & 0 \end{bmatrix}.$$

Let ϕ be the normalized solution of the refinement equation (1.1) with mask a and dilation matrix M given as above. Then ϕ is a compactly supported continuous function on $\mathbb{R}^2$, and the optimal approximation order provided by $\mathbb{S}(\phi)$ is 4.

Let us verify that a satisfies the sum rules of order 4. We observe that $\alpha = (\alpha_1, \alpha_2)$ lies in $M\mathbb{Z}^2$ if and only if $\alpha_1 + \alpha_2$ is an even integer. Hence the sum rule for a polynomial p of two variables reads as follows:

$$\sum_{\alpha_1+\alpha_2\in 2\mathbb{Z}} p(\alpha)a(\alpha) = \sum_{\beta_1+\beta_2\notin 2\mathbb{Z}} p(\beta)a(\beta),$$

that is,

$$32\,p(0,0) = 10 \sum_{|\alpha_1|+|\alpha_2|=1} p(\alpha_1,\alpha_2) - \sum_{|\alpha_1|+|\alpha_2|=3} p(\alpha_1,\alpha_2).$$

We can easily verify that this condition is satisfied for all $p \in \Pi_3$, but it is not satisfied for the monomial p given by $p(x_1, x_2) = x_1^2 x_2^2$, $(x_1, x_2) \in \mathbb{R}^2$. Therefore the refinement mask a satisfies the sum rules of order 4, but not of order 5.

In the present case, $\Omega := \{(0,0), (1,0)\}$ is a complete set of representatives of the distinct cosets of $\mathbb{Z}^2/M^T\mathbb{Z}^2$. We have $2\pi(M^T)^{-1}\Omega = \{(0,0), (\pi,\pi)\}$. Since $\hat{\phi}(0,0) = 1$, in order to verify the condition $N(\phi) \cap (2\pi(M^T)^{-1}\Omega) = \emptyset$, it suffices to show that $\hat{\phi}(\pi,\pi) \neq 0$. For this purpose, we observe that

$$\hat{\phi}(\xi) = \prod_{k=1}^\infty H((M^T)^{-k}\xi), \qquad \xi \in \mathbb{R}^2,$$

where

$$H(\xi) = [32 + 20(\cos \xi_1 + \cos \xi_2) - 4 \cos (2\xi_1 + \xi_2) - 4 \cos (\xi_1 + 2\xi_2)]/64, \\ \xi = (\xi_1, \xi_2) \in \mathbb{R}^2.$$

We have $(M^T)^{-1}(\pi,\pi)^T = (0,\pi)^T$ and $H(0,\pi) > 0$. Suppose

$$(\eta_1,\eta_2)^T = (M^T)^{-k}(\pi,\pi)^T$$

for some integer $k \geq 2$. Then $|\eta_1| \leq \pi/2$ and $|\eta_2| \leq \pi/2$, so $H(\eta_1,\eta_2) > 0$. It follows that $\hat{\phi}(\pi,\pi) \neq 0$. Consequently, the exact accuracy of ϕ is 4.

By using the methods in [12], we can easily prove that the subdivision scheme associated with mask a and dilation matrix M converges uniformly. Consequently, ϕ is a continuous function. We conclude that the optimal approximation order provided by $\mathbb{S}(\phi)$ is 4.

5. THE SUBDIVISION AND TRANSITION OPERATORS

We introduce two linear operators associated with a refinement equation. One is the subdivision operator, and the other is the transition operator. When the dilation matrix M is 2 times the identity matrix, the spectral properties of the subdivision and transition operators were studied in [10] and [18]. In this section, we extend the study to the case in which M is a general dilation matrix.

Let X and Y be two linear spaces, and T a linear mapping from X to Y . The **kernel** of T , denoted by $\ker(T)$, is the subspace of X consisting of all $x \in X$ such that $Tx = 0$.

Let a be an element in $\ell_0(\mathbb{Z}^s)$ and let M be a dilation matrix. The **subdivision operator** S_a is the linear operator on $\ell(\mathbb{Z}^s)$ defined by

$$S_a u(\alpha) := \sum_{\beta \in \mathbb{Z}^s} a(\alpha - M\beta)u(\beta), \quad \alpha \in \mathbb{Z}^s,$$

where $u \in \ell(\mathbb{Z}^s)$. The **transition operator** T_a is the linear operator on $\ell_0(\mathbb{Z}^s)$ defined by

$$T_a v(\alpha) := \sum_{\beta \in \mathbb{Z}^s} a(M\alpha - \beta)v(\beta), \quad \alpha \in \mathbb{Z}^s,$$

where $v \in \ell_0(\mathbb{Z}^s)$.

The following theorem shows that the subdivision operator S_a and the transition operator T_a have the same nonzero eigenvalues. We use I and I_0 to denote the identity mapping on $\ell(\mathbb{Z}^s)$ and $\ell_0(\mathbb{Z}^s)$, respectively.

Theorem 5.1. *The transition operator T_a has only finitely many nonzero eigenvalues. For $\sigma \in \mathbb{C} \setminus \{0\}$, the linear spaces $\ker(S_a - \sigma I)$ and $\ker(T_a - \sigma I_0)$ have the same dimension. In particular, σ is an eigenvalue of S_a if and only if it is an eigenvalue of T_a .*

Proof. For $N = 1, 2, \dots$, let E_N denote the cube $[-N, N]^s$. Choose N such that E_{N-1} contains $\text{supp } a := \{\alpha \in \mathbb{Z}^s : a(\alpha) \neq 0\}$. Let $K := \sum_{n=1}^{\infty} M^{-n} E_N$. In other words, x belongs to K if and only if $x = \sum_{n=1}^{\infty} M^{-n} y_n$ for some sequence of elements $y_n \in E_N$. Let $\ell(K)$ denote the linear space of all (finite) sequences on $K \cap \mathbb{Z}^s$. Consider the linear mapping A on $\ell(K)$ given by

$$Av(\alpha) := \sum_{\beta \in K \cap \mathbb{Z}^s} a(M\alpha - \beta)v(\beta), \quad \alpha \in K \cap \mathbb{Z}^s,$$

where $v \in \ell(K)$. The dual mapping A' of A is given by

$$A'u(\beta) := \sum_{\alpha \in K \cap \mathbb{Z}^s} u(\alpha)a(M\alpha - \beta), \quad \beta \in K \cap \mathbb{Z}^s,$$

where $u \in \ell(K)$. Let I_K denote the identity mapping on $\ell(K)$. Since $\ell(K)$ is finite dimensional, we have

$$\dim(\ker(A - \sigma I_K)) = \dim(\ker(A' - \sigma I_K)).$$

Thus, in order to establish the theorem, it suffices to prove the following two relations:

$$(5.1) \quad \dim(\ker(T_a - \sigma I_0)) = \dim(\ker(A - \sigma I_K))$$

and

$$(5.2) \quad \dim(\ker(S_a - \sigma I)) = \dim(\ker(A' - \sigma I_K)).$$

For this purpose, we introduce the sets K_j ($j = 0, 1, \dots$) as follows:

$$K_j := M^{j-1} E_1 + \dots + E_1 + K.$$

In particular, $K_0 = K$. Evidently, $K_j \subseteq K_{j+1}$ for $j = 0, 1, \dots$, and $\mathbb{R}^s = \bigcup_{j=0}^{\infty} K_j$. Moreover,

$$(5.3) \quad M^{-1}(K_j + \text{supp } a) \subseteq K_{j-1}, \quad j = 1, 2, \dots$$

Indeed, we have $M^{-1}K + M^{-1}E_N = K$, and hence

$$\begin{aligned} M^{-1}(K_j + \text{supp } a) &\subseteq M^{j-2}_j E_1 + \cdots + E_1 + M^{-1}E_1 + M^{-1}K + M^{-1}E_{N-1} \\ &\subseteq K_{j-1}. \end{aligned}$$

Suppose $\sigma \neq 0$ and $v \in \ker(T_a - \sigma I_0)$. Then $\text{supp } v \subseteq K_j$ for some $j \geq 1$. We observe that $T_a v(\alpha) \neq 0$ implies $M\alpha - \beta \in \text{supp } a$ for some $\beta \in K_j$. It follows that $\alpha \in M^{-1}(\text{supp } a + K_j) \subseteq K_{j-1}$, by (5.3). In other words, $\text{supp}(T_a v) \subseteq K_{j-1}$. Using this relation repeatedly, we obtain $\text{supp}(T^j_a v) \subseteq K$. But $v = T_a v / \sigma = (T^j_a v) / \sigma^j$. Therefore, $\text{supp } v \subseteq K$, and $v|_{K \cap \mathbb{Z}^s}$ belongs to $\ker(A - \sigma I_K)$. This shows that the restriction mapping $P : v \mapsto v|_{K \cap \mathbb{Z}^s}$ maps $\ker(T_a - \sigma I_0)$ to $\ker(A - \sigma I_K)$. Moreover, $v|_{K \cap \mathbb{Z}^s} = 0$ implies $v = 0$. So P is one-to-one. Let us show that P is also onto. Suppose $Aw = \sigma w$ for some $w \in \ell(K)$. Define $v(\alpha) := w(\alpha)$ for $\alpha \in K \cap \mathbb{Z}^s$ and $v(\alpha) := 0$ for $\alpha \in \mathbb{Z}^s \setminus K$. Then $T_a v = \sigma v$. Thus, P is one-to-one and onto, thereby establishing (5.1).

In order to prove (5.2), we consider the mapping $Q : u \mapsto u^*|_{K \cap \mathbb{Z}^s}$, where u^* is the sequence given by $u^*(\alpha) := u(-\alpha)$, $\alpha \in \mathbb{Z}^s$. Suppose $u \in \ker(S_a - \sigma I)$. Then

$$u(\alpha) = \frac{1}{\sigma} \sum_{\beta \in \mathbb{Z}^s} a(\alpha - M\beta)u(\beta), \quad \alpha \in \mathbb{Z}^s.$$

It follows that

$$u^*(\alpha) = \frac{1}{\sigma} \sum_{\beta \in \mathbb{Z}^s} u^*(\beta)a(M\beta - \alpha), \quad \alpha \in \mathbb{Z}^s.$$

For $\alpha \in K_j$ ($j \geq 1$), $a(M\beta - \alpha) \neq 0$ only if $\beta \in M^{-1}(\text{supp } a + K_j) \subseteq K_{j-1}$. Hence

$$(5.4) \quad u^*(\alpha) = \frac{1}{\sigma} \sum_{\beta \in K_{j-1} \cap \mathbb{Z}^s} u^*(\beta)a(M\beta - \alpha) \quad \text{for } \alpha \in K_j \cap \mathbb{Z}^s.$$

This shows that $u^*|_{K \cap \mathbb{Z}^s}$ belongs to $\ker(A' - \sigma I_K)$. Thus, Q maps $\ker(S_a - \sigma I)$ to $\ker(A' - \sigma I_K)$. Moreover, if $u^*(\alpha) = 0$ for $\alpha \in K \cap \mathbb{Z}^s$, then it follows from (5.4) that $u^*(\alpha) = 0$ for $\alpha \in K_j \cap \mathbb{Z}^s$, $j = 1, 2, \dots$. But $\mathbb{R}^s = \bigcup_{j=1}^\infty K_j$; hence $u^*(\alpha) = 0$ for all $\alpha \in \mathbb{Z}^s$. Thus, the mapping Q is one-to-one. It is also onto. Indeed, if $w \in \ker(A' - \sigma I_K)$, then

$$w(\alpha) = \frac{1}{\sigma} \sum_{\beta \in K \cap \mathbb{Z}^s} w(\beta)a(M\beta - \alpha), \quad \alpha \in K \cap \mathbb{Z}^s.$$

For $\alpha \in K \cap \mathbb{Z}^s$, let $u^*(\alpha) := w(\alpha)$; for $\alpha \in (K_j \setminus K_{j-1}) \cap \mathbb{Z}^s$ ($j = 1, 2, \dots$), let $u^*(\alpha)$ be determined recursively by (5.4). Then $u \in \ker(S_a - \sigma I)$ and $Qu = w$. Thus, Q is one-to-one and onto, so that (5.2) is valid. The proof of the theorem is complete. $\square$

A sequence u on $\mathbb{Z}^s$ is called a **polynomial sequence** if there exists a polynomial p such that $u(\alpha) = p(\alpha)$ for all $\alpha \in \mathbb{Z}^s$. The degree of u is the same as the degree of p . For a nonnegative integer k , let P_k be the linear space of all polynomial sequences of degree at most k , and let

$$V_k := \left\{ v \in \ell_0(\mathbb{Z}^s) : \sum_{\alpha \in \mathbb{Z}^s} p(\alpha)v(\alpha) = 0 \ \forall p \in \Pi_k \right\}.$$

For $u \in \ell(\mathbb{Z}^s)$ and $v \in \ell_0(\mathbb{Z}^s)$, we define

$$\langle u, v \rangle := \sum_{\alpha \in \mathbb{Z}^s} u(\alpha)v(\alpha).$$

Theorem 5.2. *Let M be an $s \times s$ dilation matrix and Ω a complete set of representatives of the distinct cosets of $\mathbb{Z}^s/M^T\mathbb{Z}^s$. For any $a \in \ell_0(\mathbb{Z}^s)$, the following statements are equivalent:*

- (a) *The sequence a satisfies the sum rules of order $k + 1$.*
- (b) *V_k is invariant under the transition operator T_a .*
- (c) *P_k is invariant under the subdivision operator S_a .*
- (d) *$D^\mu H(2\pi(M^T)^{-1}\omega) = 0$ for all $|\mu| \leq k$ and all $\omega \in \Omega \setminus \{0\}$.*

Proof. (a) $\Rightarrow$ (b): Let $p \in \Pi_k$ and $v \in V_k$. We have

$$\sum_{\alpha \in \mathbb{Z}^s} p(\alpha)T_a v(\alpha) = \sum_{\beta \in \mathbb{Z}^s} \left[\sum_{\alpha \in \mathbb{Z}^s} p(\alpha)a(M\alpha - \beta) \right] v(\beta).$$

Let $q(x) := p(M^{-1}x)$, $x \in \mathbb{R}^s$. Then $p(x) = q(Mx)$, $x \in \mathbb{R}^s$. By Taylor’s formula, we have

$$q(M\alpha) = q(M\alpha - \beta + \beta) = \sum_{|\mu| \leq k} q_\mu(M\alpha - \beta)\beta^\mu,$$

where $q_\mu := D^\mu q/\mu! \in \Pi_k$. Hence

$$\sum_{\alpha \in \mathbb{Z}^s} p(\alpha)a(M\alpha - \beta) = \sum_{\alpha \in \mathbb{Z}^s} q(M\alpha)a(M\alpha - \beta) = \sum_{|\mu| \leq k} c_\mu \beta^\mu,$$

where

$$c_\mu := \sum_{\alpha \in \mathbb{Z}^s} q_\mu(M\alpha - \beta)a(M\alpha - \beta)$$

is independent of β , by condition (a). Thus, we obtain

$$\sum_{\alpha \in \mathbb{Z}^s} p(\alpha)T_a v(\alpha) = \sum_{|\mu| \leq k} c_\mu \sum_{\beta \in \mathbb{Z}^s} \beta^\mu v(\beta) = 0,$$

because $v \in V_k$. This shows that $T_a v \in V_k$ for $v \in V_k$. In other words, V_k is invariant under T_a .

(b) $\Rightarrow$ (c): Suppose $p \in P_k$. We wish to show that $u := S_a p$ lies in P_k . We claim that $\langle u, v \rangle = 0$ for all $v \in V_k$. Indeed,

$$\begin{aligned} \langle u, v \rangle &= \sum_{\alpha \in \mathbb{Z}^s} u(\alpha)v(\alpha) = \sum_{\alpha \in \mathbb{Z}^s} \sum_{\beta \in \mathbb{Z}^s} a(\alpha - M\beta)p(\beta)v(\alpha) \\ &= \sum_{\beta \in \mathbb{Z}^s} p(-\beta) \sum_{\alpha \in \mathbb{Z}^s} a(M\beta - \alpha)v(-\alpha) = \sum_{\beta \in \mathbb{Z}^s} p(-\beta)w(\beta), \end{aligned}$$

where $w := T_a v^*$ with v^* given by $v^*(\alpha) = v(-\alpha)$, $\alpha \in \mathbb{Z}^s$. Since V_k is invariant under T_a and $v^* \in V_k$, we have $w \in V_k$. It follows that

$$\langle u, v \rangle = \sum_{\beta \in \mathbb{Z}^s} p(-\beta)w(\beta) = 0.$$

For a multi-index μ with $|\mu| = k + 1$, we have $\nabla^\mu \delta_\alpha \in V_k$ for all $\alpha \in \mathbb{Z}^s$. Hence $\langle u, \nabla^\mu \delta_\alpha \rangle = 0$. In other words, $\nabla^\mu u(\alpha) = 0$ for all $\alpha \in \mathbb{Z}^s$ and $|\mu| = k + 1$. This shows that u is a polynomial sequence of degree at most k .

(c) $\Rightarrow$ (a): For $p \in \Pi_k$, let $q(\gamma) := \sum_{\beta \in \mathbb{Z}^s} a(M\beta + \gamma) p(M\beta + \gamma)$ for $\gamma \in \mathbb{Z}^s$. We claim that q is a polynomial sequence. Indeed, by using Taylor's formula, we have

$$\dot{p}(M\beta + \gamma) = \sum_{|\mu| \leq k} t_\mu(M\beta) \gamma^\mu,$$

where $t_\mu := D^\mu p / \mu!$. Set $q_\mu(\beta) := t_\mu(-M\beta)$ for $\beta \in \mathbb{Z}^s$. Then for $\gamma \in \mathbb{Z}^s$,

$$\begin{aligned} q(\gamma) &= \sum_{\beta \in \mathbb{Z}^s} a(M\beta + \gamma) p(M\beta + \gamma) \\ &= \sum_{\beta \in \mathbb{Z}^s} \sum_{|\mu| \leq k} a(\gamma + M\beta) q_\mu(-\beta) \gamma^\mu = \sum_{|\mu| \leq k} (S_a q_\mu)(\gamma) \gamma^\mu. \end{aligned}$$

Note that q_μ is a polynomial sequence of degree at most k . By condition (c), $S_a q_\mu$ is a polynomial sequence; hence so is q . We observe that $q(\gamma + M\eta) = q(\gamma)$ for all $\eta \in \mathbb{Z}^s$ and $\gamma \in \mathbb{Z}^s$, that is, q is a constant sequence on the lattice $\gamma + M\mathbb{Z}^s$ for each $\gamma \in \mathbb{Z}^s$. Hence q itself must be a constant sequence. This verifies condition (a).

Finally, the equivalence between (a) and (d) was proved in Lemma 3.3. $\square$

We remark that the equivalence between (c) and (d) was proved in [7, p. 98] for the case when the dilation matrix M is 2 times the identity matrix.

6. SMOOTHNESS AND APPROXIMATION ORDER

In this section we discuss the relationship between approximation and smoothness properties of a refinable function.

Suppose ϕ satisfies the refinement equation (1.1) with the dilation matrix M being 2 times the identity matrix. It was proved by Jia in [18] that $\phi \in W_1^k(\mathbb{R}^s)$ and $\hat{\phi}(0) \neq 0$ imply that $\Pi_k \subset \mathbb{S}(\phi)$ and $\mathbb{S}(\phi)$ provides approximation order $k + 1$. This result improves an earlier result of Cavaretta, Dahmen, and Micchelli about polynomial reproducibility of smooth refinable functions (see [7, p. 158]).

The above results can be extended to the case in which the dilation matrix is isotropic. Let M be an $s \times s$ matrix with its entries in $\mathbb{C}$. We say that M is **isotropic** if M is similar to a diagonal matrix $\text{diag} \{ \lambda_1, \dots, \lambda_s \}$ with $|\lambda_1| = \dots = |\lambda_s|$. For example, for $a, b \in \mathbb{R}$, the matrix

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

is isotropic. Obviously, a matrix M is isotropic if and only if its transpose M^T is isotropic.

Lemma 6.1. *Let M be an isotropic matrix with spectral radius σ . For any vector norm $\| \cdot \|$ on $\mathbb{R}^s$, there exist two positive constants C_1 and C_2 such that the inequalities*

$$C_1 \sigma^n \|v\| \leq \|M^n v\| \leq C_2 \sigma^n \|v\|$$

hold true for every positive integer n and every vector $v \in \mathbb{R}^s$.

Proof. Since M is isotropic, we can find a basis $\{v_1, \dots, v_s\}$ for $\mathbb{C}^s$ such that $Mv_j = \lambda_j v_j$ with $|\lambda_1| = \dots = |\lambda_s| = \sigma$. Recall that two norms on a finite-dimensional linear space are equivalent. Hence there exist two positive constants C_1 and C_2 such that

$$C_1 \sum_{j=1}^s |a_j| \leq \|v\| \leq C_2 \sum_{j=1}^s |a_j| \quad \text{for } v = \sum_{j=1}^s a_j v_j.$$

But for $v = \sum_{j=1}^s a_j v_j$ we have $M^n v = \sum_{j=1}^s a_j \lambda_j^n v_j$. It follows that

$$\|M^n v\| \leq C_2 \sum_{j=1}^s |a_j \lambda_j^n| = C_2 \sigma^n \sum_{j=1}^s |a_j| \leq C_2 C_1^{-1} \sigma^n \|v\|$$

and

$$\|M^n v\| \geq C_1 \sum_{j=1}^s |a_j \lambda_j^n| = C_1 \sigma^n \sum_{j=1}^s |a_j| \geq C_1 C_2^{-1} \sigma^n \|v\|.$$

This completes the proof of the lemma.

Lemma 6.2. *Let M be an isotropic matrix with spectral radius σ . For an infinitely differentiable function f on $\mathbb{R}^s$, let*

$$f_n(\xi) := f((M^T)^n \xi), \quad \xi \in \mathbb{R}^s, \quad n = 0, 1, 2, \dots.$$

Then, for each positive integer r , there exists a positive constant C depending only on r and the matrix M such that

$$(6.1) \quad \max_{|\mu|=r} |D^\mu f_n(\xi)| \leq C \sigma^{rn} \max_{|\nu|=r} |D^\nu f((M^T)^n \xi)| \quad \forall \xi \in \mathbb{R}^s.$$

Proof. Let $B = (b_{pq})_{1 \leq p, q \leq s}$ be the matrix $(M^T)^n$. By the chain rule, for $j = 1, \dots, s$, we have

$$D_j f_n(\xi) = (b_{1j} D_1 + \dots + b_{sj} D_s) f((M^T)^n \xi), \quad \xi \in \mathbb{R}^s.$$

Hence, for a multi-index $\mu = (\mu_1, \dots, \mu_s)$ with $|\mu| = r$,

$$D^\mu f_n(\xi) = \prod_{j=1}^s D_j^{\mu_j} f_n(\xi) = \prod_{j=1}^s (b_{1j} D_1 + \dots + b_{sj} D_s)^{\mu_j} f((M^T)^n \xi), \quad \xi \in \mathbb{R}^s.$$

By Lemma 6.1, there exists a constant $C_1 > 0$ depending only on the matrix M such that $|b_{pq}| \leq C_1 \sigma^n$ for all p, q . We may express $\prod_{j=1}^s (b_{1j} D_1 + \dots + b_{sj} D_s)^{\mu_j}$ as $\sum_{|\nu|=r} c_\nu D^\nu$, where each c_ν is a linear combination of products of r factors of the b_{pq} 's. Hence there exists a positive constant C depending only on r and the matrix M such that $|c_\nu| \leq C \sigma^{rn}$ for all $|\nu| = r$. This proves (6.1). $\square$

Now we are in a position to establish the main result of this section.

Theorem 6.3. *Suppose M is an $s \times s$ isotropic dilation matrix, and a is an element in $\ell_0(\mathbb{Z}^s)$ satisfying (1.2). Let ϕ be the normalized solution of the refinement equation (1.1). If $\phi \in W_1^k(\mathbb{R}^s)$, then $\Pi_k \subset \mathbb{S}(\phi)$ and $\mathbb{S}(\phi)$ provides approximation order $k + 1$.*

Proof. Since $\hat{\phi}(0) = 1$, in order to prove $\mathbb{S}(\phi) \supset \Pi_k$, it suffices to show that for $|\mu| \leq k$,

$$(6.2) \quad D^\mu \hat{\phi}(2\pi\beta) = 0 \quad \forall \beta \in \mathbb{Z}^s \setminus \{0\}.$$

The proof proceeds with induction on $|\mu|$, the length of μ .

Let H be the function given in (3.2). A repeated application of (3.1) yields that, for $n = 1, 2, \dots$,

$$\hat{\phi}(\xi) = \left[\prod_{j=1}^n H((M^T)^{-j} \xi) \right] \hat{\phi}((M^T)^{-n} \xi), \quad \xi \in \mathbb{R}^s.$$

It follows that

$$(6.3) \qquad \hat{\phi}((M^T)^n \xi) = h_n(\xi) \hat{\phi}(\xi), \qquad \xi \in \mathbb{R}^s,$$

where $h_n(\xi) := \prod_{j=1}^n H((M^T)^{j-1} \xi)$. Note that H is 2π -periodic and $H(0) = 1$. Thus, we have

$$\hat{\phi}(2\pi(M^T)^n \beta) = \left[\prod_{j=1}^n H(2\pi(M^T)^{j-1} \beta) \right] \hat{\phi}(2\beta\pi) = \hat{\phi}(2\beta\pi), \qquad \beta \in \mathbb{Z}^s.$$

If $\phi \in L_1(\mathbb{R}^s)$, then by the Riemann-Lebesgue lemma we obtain

$$\hat{\phi}(2\beta\pi) = \lim_{n \rightarrow \infty} \hat{\phi}(2\pi(M^T)^n \beta) = 0 \qquad \forall \beta \in \mathbb{Z}^s \setminus \{0\}.$$

This establishes (6.2) for $\mu = 0$.

Let $0 < r \leq k$. Assume that (6.2) has been proved for $|\mu| < r$. We wish to establish (6.2) for $|\mu| = r$. For this purpose, we deduce from (6.3) that

$$\hat{\phi}(\xi) = f_n(\xi) [1/h_n(\xi)], \qquad \xi \in \mathbb{R}^s,$$

where $f_n(\xi) := \hat{\phi}((M^T)^n \xi)$, $\xi \in \mathbb{R}^s$. By using the Leibniz formula for differentiation, we get

$$(6.4) \qquad D^\mu \hat{\phi}(\xi) = \sum_{\nu \leq \mu} \binom{\mu}{\nu} D^\nu f_n(\xi) D^{\mu-\nu} [1/h_n](\xi), \qquad \xi \in \mathbb{R}^s.$$

But, for $\beta \in \mathbb{Z}^s \setminus \{0\}$ and $|\nu| < r$, we have $D^\nu f_n(2\pi\beta) = 0$, by the induction hypothesis. When $\nu = \mu$, we have $[1/h_n](2\pi\beta) = 1$. Hence it follows from (6.4) that

$$(6.5) \qquad D^\mu \hat{\phi}(2\pi\beta) = D^\mu f_n(2\pi\beta), \qquad \beta \in \mathbb{Z}^s \setminus \{0\}.$$

By Lemma 6.2, we have

$$(6.6) \qquad |D^\mu f_n(2\pi\beta)| \leq C \sigma^{rn} \max_{|\nu|=r} |D^\nu \hat{\phi}((M^T)^n 2\pi\beta)|, \qquad \beta \in \mathbb{Z}^s \setminus \{0\},$$

where $C > 0$ is a constant independent of n .

In what follows, we use v_j to denote the j th coordinate of a vector v in $\mathbb{R}^s$. For a multi-index $\nu = (\nu_1, \dots, \nu_s)$, let ϕ_ν be the function given by $\phi_\nu(x) = (-ix)^\nu \phi(x)$, $x \in \mathbb{R}^s$. Then $D^\nu \hat{\phi} = \hat{\phi}_\nu$ and

$$((-iD_j)^r \phi_\nu)^\wedge(\xi) = \xi_j^r D^\nu \hat{\phi}(\xi), \qquad \xi = (\xi_1, \dots, \xi_s) \in \mathbb{R}^s.$$

Since $\phi \in W_1^k(\mathbb{R}^s)$, we have $(-iD_j)^r \phi_\nu \in L_1(\mathbb{R}^s)$. Thus, by the Riemann-Lebesgue lemma, we obtain

$$\lim_{n \rightarrow \infty} ((M^T)^n \beta)_j^r D^\nu \hat{\phi}(2\pi(M^T)^n \beta) = 0 \qquad \text{for } \beta \in \mathbb{Z}^s \setminus \{0\}.$$

This is true for $j = 1, \dots, s$; hence it follows that

$$\lim_{n \rightarrow \infty} \|(M^T)^n \beta\|^r D^\nu \hat{\phi}(2\pi(M^T)^n \beta) = 0 \qquad \text{for } \beta \in \mathbb{Z}^s \setminus \{0\},$$

where $\|\cdot\|$ is a vector norm on $\mathbb{R}^s$. By Lemma 6.1, there exists a positive constant $C_1 > 0$ independent of n such that

$$C_1 \sigma^n \|\beta\| \leq \|(M^T)^n \beta\|.$$

Therefore

$$\lim_{n \rightarrow \infty} \sigma^{nr} D^\nu \hat{\phi}(2\pi(M^T)^n \beta) = 0 \quad \text{for } \beta \in \mathbb{Z}^s \setminus \{0\}.$$

This in connection with (6.5) and (6.6) tells us that $D^\mu \hat{\phi}(2\pi\beta) = 0$ for $|\mu| = r$ and $\beta \in \mathbb{Z}^s \setminus \{0\}$. The proof of the theorem is complete. $\square$

Recall that Ω is a complete set of representatives of the distinct cosets of $\mathbb{Z}^s/M^T\mathbb{Z}^s$. Thus, as a consequence of Theorem 6.3, we conclude that if the normalized solution ϕ of the refinement equation (1.1) lies in $W_1^k(\mathbb{R}^s)$, and if $N(\phi) \cap (2\pi(M^T)^{-1}\Omega) = \emptyset$, then the refinement mask a satisfies all the conditions in Theorem 5.2.

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